

Research Article

Applying ChatGPT and AI-Powered Tools to Accelerate Evidence Reviews

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Artificial intelligence (AI) tools have been used to improve the productivity of evidence review and synthesis since at least 2016, with EPPI-Reviewer and Abstrackr being two prominent examples. However, since the release of ChatGPT by OpenAI in late 2022, the use of generative AI for research, especially for text-based data analysis, has exploded. In this article, we used a critical reflection approach to document and evaluate the capacity of different generative AI tools such as ChatGPT, GPT for Google Sheets and Docs, Casper AI, and ChatPDF to assist in the early stages of a rapid evidence review process. Our results demonstrate that these tools can boost research productivity in formulating search strings and screening literature, but they have some notable weaknesses, including producing inconsistent results and occasional errors. We recommend that researchers exercise caution when using generative AI technologies by designing a thorough research strategy and review protocol to ensure effective monitoring and quality control.

Keywords: Artificial intelligence; ChatGPT; evidence synthesis; large language models; rapid review; research methods

1. Introduction

The release of ChatGPT, a generative pretrained transformer (GPT), as a public tool by OpenAI in November 2022 represented a major advancement in artificial intelligence (AI) technologies [1], particularly in the realm of text-based data analysis. ChatGPT, as a large language model (LLM) with the latest model being GPT-4o released in May 2024, sets itself apart from earlier machine learning models by its ability to understand user requests and produce personalized, human-like responses, potentially impacting multiple industries [2]. Harnessing the power of natural language processing (NLP), computer vision, big data, and deep learning, coupled with its intuitive chatbot interface, ChatGPT and other generative AI have the potential to transform many areas such as climate change [3], public health [4], and especially education [5]. Capable of understanding natural lan-

guage, ChatGPT can analyze qualitative unstructured data, suggesting codes, formulating codebooks and categories, and generating themes [6–8].

In the field of evidence-based research, the proliferation of Web pages has posed natural and social sciences with a “data deluge” challenge [9]. This challenge intensifies with the growing volume of online accessed databases of published material, which requires the application of “powerful new technologies such as text mining” to tackle it ([10], p. 522). Accordingly, early AI and machine learning tools, such as EPPI-Reviewer and Abstrackr, have assisted with the time-consuming tasks of title and abstract screening in evidence review and synthesis at least since 2016 [11, 12]. In their scoping review of different tools in automating systematic reviews including LitSuggest, Rayyan, Abstrackr, BIBOT, R software, RobotAnalyst, DistillerSR, ExaCT, and NetMetaXL, [13] found that these tools, while having the potential to support systematic reviews,

remain somewhat user-unfriendly due to the necessity for users to have expertise in computer science and algorithms. COVID-19, as an unprecedented global pandemic and crisis, “created a sudden, urgent and shared imperative” for knowledge synthesis in order to more urgently inform policy and decision-making ([14], p.9). In this situation, the emergence of ChatGPT and other generative AI tools offers a significant opportunity to automate resource-heavy tasks in systematic literature reviews, enabling the production of evidence with efficiency, affordability, transparency, and rigour [15]. Moreover, with their friendly user interfaces and language capabilities, ChatGPT and AI chatbots are accessible to many users, including those without a background in computer sciences or sufficient English proficiency. AI chatbots can “understand” users’ prompts even if they are written in everyday language or contain grammatical and spelling errors. It has been evident that these AI-powered tools are capable of assisting in and improving the efficiency and transparency of systematic reviews through different stages, from the topic refinement phase to the knowledge synthesis phase [16, 17]. Wang et al. [18] experimented on over 100 systematic review topics and found that ChatGPT could effectively produce Boolean queries when guided by carefully designed prompts. Despite the expansion of this scholarship concerning this topic, the adoption of new AI tools such as ChatPDF, Casper AI, GPT for Google Sheets and Docs, and ChatGPT in the literature review is still comparatively immature [17]. Therefore, further trials and evaluations of AI tools’ strengths and weaknesses in the process of evidence reviews are required.

In this article, we explore the use of several AI-powered tools based on LLMs, including ChatGPT [19], GPT for Google Sheets and Docs [20], Casper AI [21], and ChatPDF [22], to assist in the early stages of the evidence review process. Table 1 outlines the key features of these four AI tools. The purpose of this article is twofold: first, to showcase the potential of AI-powered tools for evidence review in research and academia, and second, to identify their current weaknesses and provide recommendations for researchers looking to utilize them.

2. Data and Methods

2.1. Outline of a Rapid Review Methodology With AI-Powered Tools. Rapid review, rapid evidence assessment, or rapid systematic review is “a form of knowledge synthesis that accelerates the process of conducting a traditional systematic review through streamlining or omitting a variety of methods to produce evidence in a resource-efficient manner” ([23], p.395). Although adhering to the standards of systematic reviews, rapid reviews are designed specifically to be resource efficient by addressing the challenges of financial and human resource shortages, time constraints, and urgent and emergent demands of decision-making in a shorter period [23, 24]. As such, rapid reviews are often narrowed down by the types of evidence (e.g., focusing on review papers and excluding grey literature), the number of databases (e.g., including only two databases), and the number of reviewers involved throughout the review process. In this experiment, we argue that with the availability of ChatGPT and generative AI tools, there is a possibility of

improving the efficiency of rapid reviews in times of resource shortage.

In this article, based on the practical guidelines for rapid review outlined in Bragge et al. [25] and recommendations from Speckemeier et al. [23], we designed and employed the eight basic steps of a rapid review as shown in Figure 1: need assessment, developing a research protocol, literature search, screening and study selection, data extraction, quality assessment, knowledge synthesis, and reporting findings. Our focus was to apply AI tools in Steps 2 and 4.

Step 2 involves developing a review protocol that comprises the research question, database search string(s), inclusion and exclusion criteria, and method for searching, screening, extracting, and synthesizing findings (see [23]). Our main focus in this step was on using ChatGPT to support the development of search strings (Boolean queries) and adapt them to different target databases. Moving to Step 4, we demonstrate how to use GPT for Google Sheets and Docs to screen titles and abstracts, and ChatPDF and Casper AI for full-text screening. For the remaining steps, we are of the view that LLM and other AI-powered tools can significantly enhance the execution of each step in a rapid review process. We discuss our perspective in detail towards the end of the article.

2.2. Research Topics and Case Study. To experiment with the use of ChatGPT and other AI-powered tools in evidence review, we drew on a case study (Table 2) that is aimed at better understanding the risks and opportunities associated with the participation of land managers in environmental markets in the Goulburn Broken catchment, Victoria, Australia [26]. This study’s research question and objectives are outlined below.

Understanding this information determined this evidence review’s need, topic selection and refinement, and the meaning and importance of keywords (Step 1, Figure 1), which guided the development of our review protocol (see Step 2, Results section). We did not apply AI-powered tools to this stage of the evidence review. A detailed description of the topic selection and context for the project is outside the scope of this article.

2.3. Qualitative Assessment of AI-Generated Outputs. In this experiment, our methodological approach was qualitative in nature. This means that we did not aim to employ a traditional experimental design to compare and contrast the treatment and control groups’ outcomes (e.g., comparing ChatGPT’s and human researchers’ outputs). Instead, AI performance was closely monitored and guided by our expertise in behavioral science, environmental markets and farmers’ participation, and best practice research methodology in an evidence review. In doing so, we employed critical reflection as a research method of inquiry in its own right [27–29]. Critical reflection can help improve our professional practice because it demands that we be aware of the ways we use to produce knowledge including the epistemological assumptions guiding us in doing so [30]. These questions can be incorporated in a research design that requires researchers to create their personal notes to reflect on their

TABLE 1: Comparison between four AI tools.

Feature	ChatGPT	GPT for Google Sheets and Docs	Casper AI	ChatPDF
Company	OpenAI	Talarian	Casper AI	ChatPDF GmbH
AI Model used	GPT-4	GPT-4	Custom AI model	Custom AI model
Classification	Web-based chatbot	Add-on	Google Chrome extension	Web-based chatbot and PDF reader
User interface	Chat based	Integrated with Google Sheets/Docs	Intergrated with Google Chrome, chat-based sidebar	PDF reader and chat-based interface
Data type	Text, image, and datasets (e.g., CSV)	Text and spreadsheet data	Text (Webpage) and PDF documents	PDF documents only
Text generation	Yes	Yes	Yes	Yes
Image generation and understanding	Yes (via DALL-E)	No	No	No
Customization	API available for custom applications	Customizable within Google Workspace	Customizable with limitations	API available for custom applications
Free plan	Yes (GPT-3.5 and GPT-4)	No free plan	Yes, 5 free credits per day	Yes, 2 PDFs per day, each up to 120 pages
Subscription cost	\$20/month	Pay as you use	\$7/month	\$20/month

Note: Source: authors compiled based on Casper AI [21] ChatPDF [22], OpenAI [19], and Talarian [20].

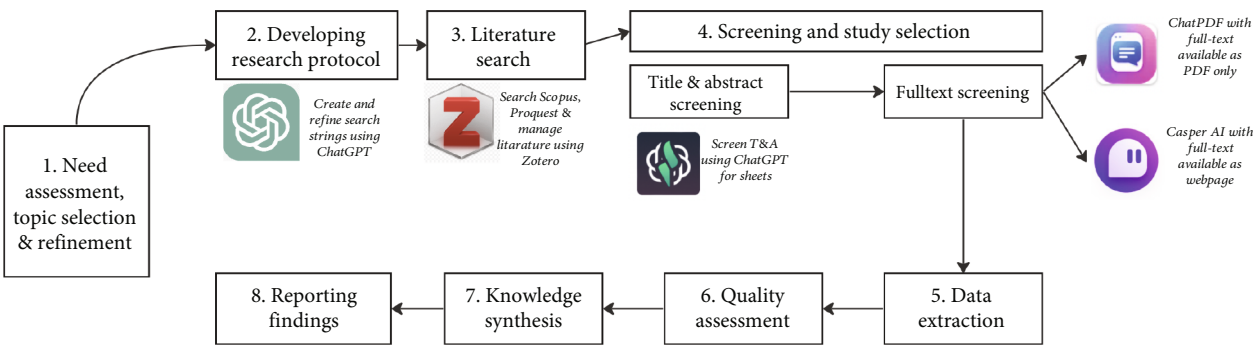


FIGURE 1: Rapid review process with the support of AI-powered tools.

TABLE 2: Research question and objectives of our case study.

Primary research question	• Does land managers’ participation in environmental markets build climate resilience? • List of environmental market participation outcomes that align with climate resilience outcomes (and those that potentially do not)
Research objectives underpinning the research question	• List of determinants (drivers and barriers) of participation in environmental markets building climate resilience • List of interventions/activities that appear to be relevant and effective in ensuring environmental markets participation leads to climate resilience

experience and ask reflexive questions to facilitate in-depth discussions within the research team (see our similar approach in [31]). In the current study, critical reflection was used to assist us in critically identifying how we experimented with AI tools in evidence review. In this regard, we asked questions such as whether AI tools understand the certain concepts, what drives AI tools in generating certain outputs, and whether these outputs meet our research requirements. In this process, our expertise in environmen-

tal markets, behavioral science, and rapid evidence review methodologies was important in helping us assess AI-generated outputs and their biases. In particular, we focused on qualitatively recording (authors’ personal and collaborative notes) and critically reflecting on AI potentials and limitations based on their produced outputs and our research requirements (i.e., the research questions and scope of our research project on the environmental market and farmers’ participation). In this

TABLE 3: Summary of Boolean query development.

Role	Text	Commentary
User	Please develop a search string that helps answer the following 3 objectives. List of environmental market participation outcomes that align with climate resilience outcomes (and those that potentially do not). [...]	Initial prompt to generate a search string
ChatGPT	Sure! Please find below search strings for each of the three research objectives 1. List of environmental market participation outcomes that align with climate resilience outcomes (and those that potentially do not) ("environmental market*" OR "carbon market*" OR "carbon offset*" OR "carbon credit*" OR "cap and trade" OR "emissions trading") "climate resilience" OR [...]	
User	Could you please combine all three search strings into ONE only?	We sought to create a single string to address all objectives
ChatGPT	Sure, here is a combined search string for all three research objectives ("environmental market*" OR "carbon market*" OR "carbon offset*" OR "carbon credit*" OR "cap and trade" OR "emissions trading") "climate resilience" OR [...]	
User	Did you forget to add "AND" to the search string?	The response did not include the correct AND operators between elements of the string
ChatGPT	[...] ("environmental market*" OR "carbon market*" OR "carbon offset*" OR "carbon credit*" OR "cap and trade" OR "emissions trading") AND ("climate resilience" OR [...])	
User	Very good. But in the first objective, there is a term "participation in environmental market". Where is it in the search string?	Although the string was now much better, we still found it missing the key words "participation".
ChatGPT	[...] ("participation in environmental market*" OR "carbon market*" OR "carbon offset*" OR "carbon credit*" OR "cap and trade" OR "emissions trading") AND ("climate [...])	
User	Adapt the following search string to be used in ProQuest database ("participation in environmental market*" OR "carbon market*" OR "carbon offset*" OR "carbon [...])	As the string was now good, we asked ChatGPT to adapt this string to ProQuest. We also requested that ChatGPT add "NOFT" to every keyword. NOFT stands for "Anywhere except full text", namely, ProQuest will only search for abstract, title and keywords.
ChatGPT	noft(((noft(determinants) OR noft(drivers) OR noft(barriers) OR noft(facilitators) OR noft(challenges) OR noft(obstacles) OR noft(opportunities) OR noft(factors) OR noft(enablers) OR noft(constraints) OR [...])	
User	Adapt this search string to be used in Scopus database noft(((noft(determinants) OR noft(drivers) OR noft(barriers) OR noft(facilitators) OR noft(challenges) OR noft(obstacles) OR noft(opportunities) OR noft(factors) OR noft(enablers) OR [...])	We requested that the prompt be adapted for Scopus

TABLE 3: Continued.

Role	Text	Commentary
ChatGPT	(TITLE-ABS-KEY (determinants OR drivers OR barriers OR facilitators OR challenges OR obstacles OR opportunities OR factors OR enablers OR constraints OR motivations OR disincentives OR requirements OR conditions OR influences OR impacts) [...])	ChatGPT did a good job of adapting the string. Note that it recognised Scopus has the search function “Title-Abs-Key,” which is different from ProQuest.

Note: Source: authors.

Strengths

- ChatGPT facilitates the quick development of search strings from research objectives or questions.
- ChatGPT can assist in quickly adapting search strings from one database to another.

Weaknesses

- ChatGPT does not understand the difference in significance of keywords in the search strings. Thus, researchers need to determine which keywords are crucial for their research objectives.
- ChatGPT makes errors, such as forgetting operators, which could compromise the search results. Researchers need to be able to verify and correct these mistakes.

Box 1: Advantages and disadvantages of using ChatGPT in formulating search strings.

process, we tried to establish a feedback loop where team members constantly provided qualitative feedback on the AI-generated outputs and helped guide AI performance. In Step 2, after we had established the clear topic and research question for our evidence review, we proceeded with our experiment with ChatGPT in getting its support in crafting Boolean queries for our database search (Step 3). One author conducted this experiment with ChatGPT (model GPT-4, OpenAI [19]) and recorded the procedure and results, which included the user’s prompts and ChatGPT’s responses in both text and screenshots. We then had another author check the results and provided qualitative feedback on the quality of the Boolean query. The two authors agreed on these assessments, which were later recorded in the *Commentary* column in Table 3.

In Step 4, we organized our recording in a shared Google Sheets spreadsheet which included Scopus and ProQuest database entries (e.g., authors, title of papers, journals, and abstract note). In the process of screening, we created extra columns for GPT for Google Sheets and Docs to operate the screening job. To determine the relevance of each study, we created a group of columns containing key concepts that helped point us to the study’s topics such as “farmers/land manager,” “carbon market,” “agritourism,” “ecosystem market,” “biodiversity market,” “renewable energy market,” and “climate change.” In “farmers/land manager” column, for example, we instructed GPT for Google Sheets and Docs to scan if the study’s abstract contained in the *Abstract note* column mentioned farmers or land managers. The AI-generated answers were yes/no. In addition to these columns, we had a final column titled “Included” that allowed us to ask GPT for Google Sheets and Docs to scan the individual columns above and notify us if any of those columns contained “yes.” This procedure allowed us to determine the relevance of individual studies and if we should include

them for data extraction. This “Included” column helped reduce potential biases in GPT for Google Sheets and Docs based on its single scanning results (i.e., yes/no) in the aforementioned columns. In this process, one author experimented with all the prompts to guide GPT for Google Sheets and Docs, and the second author reviewed the results in the mentioned columns. Any disagreement was resolved by consulting the third author. Shared qualitative assessments of AI tools in each of these steps were recorded and provided in boxes in the Results section.

2.4. Study Limitations. Due to our time constraints and resources within our practical project, our experiment was only able to focus on two steps (Steps 2 and 4) of the rapid evidence review process, namely, formulating Boolean queries and title and abstract and full-text screening. Our small number of initial results returned from ProQuest and Scopus could limit our assessment of AI capacity in carrying out evidence review. Future research should expand to more databases and a larger number of citations for screening. Additionally, future studies should explore the potential of AI tools such as ChatGPT in other steps such as topic refinement, literature search (with new tools including Perplexity, Elicit, Scispace, and Consensus), and knowledge synthesis. Our qualitative approach to experimenting with AI tools in the evidence review process could make our judgment and reporting in this article subject to biases. Future work should design the experiments to compare human researchers and AI’s outputs and test these parameters.

3. Results of Using AI-Powered Tools in Rapid Evidence Review Procedure

In this section, we focus on describing the selected steps of the evidence review (Steps 2 and 4) where we utilized the

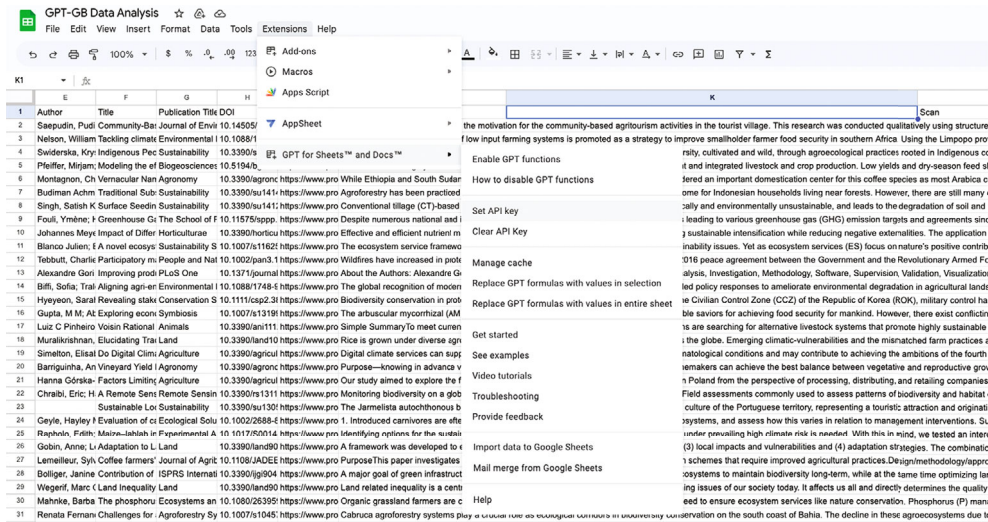


FIGURE 2: Enabling GPT functionality in a Google Sheet.

aforementioned AI tools and provided our qualitative evaluation on their potential as well as their weaknesses.

3.1. Developing a Research Protocol—Using ChatGPT. In this section (Step 2, Figure 1), we focus on the development of search strings or Boolean queries. Other elements of the research protocol, such as inclusion and exclusion criteria and methods for searching, screening, extracting, and synthesizing findings, were not developed through the use of AI.

We used the research objectives from Step 1 to develop a search string with keywords, provide feedback on the responses from ChatGPT, and format it for the two databases we intended to search (ProQuest and Scopus). A detailed exchange with prompts and responses is provided in Appendix 1 in the Supporting Information section.

Using these search strings, we conducted searches on the databases ProQuest, Scopus, and Google Scholar (Step 3 in Figure 1). On ProQuest, the string returned 56 records. After removing one non-English study, we had 55 studies for the next stage of review. On Scopus, the string returned 118 records. After limiting the search results to include only articles, reviews, book chapters, and conference papers, the final number of records was 81. On Google Scholar, the string returned 4500 hits, but we only screened the first five pages and selected 19 results. Typically, initial searches in these databases result in a larger number of hits. However, for the purposes of this experiment and the limited scope of the practical project, we accepted the small number of studies found and acknowledged this limitation in assessing AI capacity. In addition, there were three studies suggested by experts in the project interviews [26], making a total of 158 studies.

We imported the final records into Zotero—a free reference management software. After removing duplicates and non-English results (24), we ended up with a total of 134 records for title and abstract screening using AI tools. We exported the Zotero database to spreadsheet format (as a CSV, comma-separated values) so it could be manipulated in Google Sheets.

Below is a summary outlining the advantages and disadvantages of using ChatGPT at this particular stage.

3.2. Screening and Study Selection—Using GPT for Google Sheets and Docs

3.2.1. Title and Abstract Screening. This is Step 4 in our review process (Figure 1). Google Sheets allows plugins to connect with other software through APIs (application programming interfaces) (see Appendix 2 in the Supporting Information section for more information on costs). APIs are used to help create a “channel” for different software to work with each other. In this experiment, we used the plugin “GPT for Google Sheets and Docs” that allows for the use of a new formula called `=gpt()`, which communicates with a web-based GPT system similar to ChatGPT. A spreadsheet needs to be activated using an API key to get access to OpenAI’s ChatGPT models (Figure 2). In short, the purpose is to use ChatGPT within Google Sheets.

We sought to use GPT for Google Sheets and Docs to assist in determining the relevance of records to the research objectives of the project. We intended to exclude irrelevant records and conduct full-text screening on the remaining records. Because the case study project focused on understanding the involvement of land managers and farmers in environmental markets and its impact on climate outcomes, relevant records were those that included:

1. Land managers and farmers
2. Certain types of environmental markets Carbon market, ecosystem market, biodiversity market, and renewable energy market
3. Climate change or climate outcomes

To accelerate this screening process, we used GPT for Google Sheets to assess each abstract for each record and determine its relevance. The records exported from Zotero already included a column containing the study abstract.

K2	E	F	G	H	I	J	K
1	Author	Title	Publication Title	DOI	URL	Abstract Note	Scan
2	Saepudin, Puji	Community-Based	Journal of Env	10.14405/jent	https://www.pro	This research aims to determine the role of farmers/land manager	ely using structure
3	Nelson, William	Tackling climate	Environmental	10.1088/1748-6	https://www.pro	Sustainable intensification (SI) of low input farming systems is promoted as a strategy to improve smallholder farmer food security in southern Africa. Using the Limpopo pro	
4	Swiderska, Krysia	Indigenous Pae	Sustainability	10.3390/su1411	https://www.pro	The food systems and territories of Indigenous Peoples sustain much of the world's biodiversity, cultivated and wild, through agroecological practices rooted in Indigenous co	
5	Pleiffer, Mirjam	Modeling the ef	Biogeosciences	10.5194/bg-19-	https://www.pro	Smallholder farming systems in southern Africa are characterized by low-input management and integrated livestock and crop production. Low yields and dry-season feed si	
6	Montagnon, Ch	Vernacular Nan	Agronomy	10.3390/agronc	https://www.pro	While Ethiopia and South Sudan are the native habitats for Coffea arabica, Yemen is considered an important domestication center for this coffee species as most Arabica co	
7	Budiman Achm	Traditional Sub	Sustainability	10.3390/su1411	https://www.pro	Agroforestry has been practiced for decades and is undoubtedly an important source of income for Indonesian households living near forests. However, there are still many i	
8	Singh, Satish K	Surface Seedin	Sustainability	10.3390/su1411	https://www.pro	Conventional tillage (CT)-based agriculture is known to be ecologically indiscreet, economically and environmentally unsustainable, and leads to the degradation of soil and	
9	Fauli, Yvonne	Greenhouse Gi	The School of E	10.11575/nyppp	https://www.pro	Despite numerous national and international climate conferences, meetings and workshops leading to various greenhouse gas (GHG) emission targets and agreements sinc	
10	Johannes Mey	Impact of Differ	Horticulturae	10.3390/horticu	https://www.pro	Effective and efficient nutrient management is central to best-practice agriculture, facilitating sustainable intensification while reducing negative externalities. The application	
11	Blanco Julien	A novel ecosys	Sustainability	10.1007/s11622	https://www.pro	The ecosystem service framework has been instrumental in navigating local to global sustainability issues. Yet as ecosystem services (ES) focus on nature's positive contrib	
12	Tebbutt, Charle	Participatory m	People and Nat	10.1002/pan3.1	https://www.pro	Wildfires have increased in protected areas (PAs) of the Colombian Amazon following the 2016 peace agreement between the Government and the Revolutionary Armed Fo	
13	Alexandre Gori	Improving prod	PLoS One	10.1371/journal	https://www.pro	About the Authors: Alexandre Gori Maia Roles Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Supervision, Validation, Visualization	
14	Blitt, Sofia	Trail Aligned agri-en	Environmental	10.1088/1748-6	https://www.pro	The global recognition of modern agricultural practices' impact on the environment has fuelled policy responses to ameliorate environmental degradation in agricultural lands	
15	Hyeyoon, Saral	Revealing statu	Conservation	10.1111/exp2.3	https://www.pro	Biodiversity conservation in protected areas requires strict legal limitations to land use. In the Civilian Control Zone (CCZ) of the Republic of Korea (ROK), military control ha	
16	Gupta, M. M.	At Exploring econ	Symbiosis	10.1007/s13191	https://www.pro	The arbuscular mycorrhizal (AM) symbiosis has long been projected as one of the sustainable saviors for achieving food security for mankind. However, there exist conflict	
17	Luiz C. Pinheiro	Volsin Rational	Animals	10.3390/ani1111	https://www.pro	Simple SummaryTo meet current global food supply challenges, researchers and technicians are searching for alternative livestock systems that promote highly sustainable	
18	Muralikrishnan, Elucudating	Trail Land		10.3390/land10	https://www.pro	Rice is grown under diverse agro-climatic conditions and crop management regimes across the globe. Emerging climatic-vulnerabilities and the mismatched farm practices i	
19	Simelton, Elsal	Do Digital Clini	Agriculture	10.3390/agricul	https://www.pro	Digital climate services can support agricultural management decisions under uncertain climatological conditions and may contribute to achieving the ambitions of the fourth	
20	Bariguiha, An	Vineyard Yield	I Agronomy	10.3390/agronc	https://www.pro	Purpose—knowing in advance vineyard yield is a critical success factor so growers and winemakers can achieve the best balance between vegetative and reproductive gro	
21	Hanna Górnska	Factors Limiting	Agriculture	10.3390/agricul	https://www.pro	Our study aimed to explore the factors limiting the development of the organic food sector in Poland from the perspective of processing, distributing, and retailing companies	
22	Chraibi, Eric	H A Remote Senti	Remote Sensin	10.3390/rs1311	https://www.pro	Monitoring biodiversity on a global scale is a major challenge for biodiversity conservation. Field assessments commonly used to assess patterns of biodiversity and habitat i	

FIGURE 3: Using GPT for Google Sheets and Docs to screen abstracts.

In a column of the Google Sheet, we used the = gpt() formula (Figure 3)

= GPT(" write yes if the abstract mentions farmer or land manager ", [Abstract cell reference])

Researchers should conduct a thorough review of the screening results from the initial screening using GPT for Google Sheets and Docs. It is important to cross-check the outputs with the abstract of the searched study and only proceed to the next step if the = gpt() formula returns accurate results.

As ChatGPT's screening was effective, we used this same formula for all 115 records to assess the presence or absence of farmers/land managers (Figure 4).

Using the same method, we searched for "carbon market," "agritourism," "ecosystem market," "biodiversity market," "renewable energy market," and "climate change" in separate columns. It is worth noting that the researchers must be the ones who decide which keywords are important for the relevance of a study.

Once we had the initial result for each of these terms, we needed to combine these responses to decide on the relevance of the record. In a new column of the sheet titled "Included," we used the = gpt() formula again.

= GPT(" write yes if any of these cells containing yes ", [farmer cell reference climate change cell reference])

The outputs of the screening process were summarized in a new column titled "Included?" (see Figure 5, the column highlighted within the red rectangle on the sheet). This column displayed the outcome (yes/no) of the concept search for each document. Documents that did not contain the specific keywords we were looking for were marked as "no" and highlighted in grey as shown in Figure 5. By sorting the entire dataset based on the "Included?" column, we could

easily identify and remove 16 irrelevant studies from our analysis.

There are four cautions to keep in mind when developing a GPT for Google Sheets and Docs formula. First, if the formula is too complicated, GPT for Google Sheets and Docs may not perform accurately. For example, in our experiment, we wrote a formula as follows:

= GPT(" write yes if the abstract mentions either carbon market, biodiversity market, natural capital market, agritourism market, water quality market or ecosystem services ", [Abstract cell reference])

K2:K12 `=GPT("write yes if this abstract mentions farmer or land manager",J2)`

	E	F	G	H	I	J	K	
1	Author	Title	Publication Title DOI	UM	Abstract Note		farmers land manager	Scan
2	Saeudin, Pudi	Community-Bai Journal of Envis	10.14505/jem	https://www.pro	The research aims to determine the motivation for sustainable intensification (SI) of low input farming systems in the tropics. This abstract mentions farmers.		Yes, this abstract mentions farmers.	
3	Nelson, William	Tackling climate Environmental	10.1088/1748-6	https://www.pro	Sustainable intensification (SI) of low input farming systems in the tropics. This abstract mentions farmers.		Yes, this abstract mentions farmers.	
4	Swiderska, Kryn	Indigenous Pec Sustainability	10.3390/su1411	https://www.pro	The food systems and territories of Indigenous Peoples in the Amazon basin are characterized by a high degree of diversity and resilience. This abstract mentions farmers.		Yes, this abstract mentions farmers.	
5	Pfleiffer, Mirjam	Modeling the effects of agroecology	10.5194/bg-19-	https://www.pro	Smallholder farming systems in southern Africa. This abstract mentions farmers.		Yes, this abstract mentions farmers.	
6	Montagnon, Ch	Vernacular Nan Agronomy	10.3390/agronc	https://www.pro	While Ethiopia and South Sudan are the native birthplaces of agroecology, the concept has been adopted by many countries. This abstract mentions farmers.		Yes, this abstract mentions farmers.	
7	Budiman Achm	Traditional Sub Sustainability	10.3390/su1411	https://www.pro	Agroforestry has been practiced for decades in the tropics. This abstract mentions farmers.		Yes, this abstract mentions farmers.	
8	Singh, Satish K	Surface Seed Sustainability	10.3390/su1411	https://www.pro	Conventional tillage (CT)-based agriculture is a major contributor to soil degradation. This abstract mentions farmers.		Yes, this abstract mentions farmers.	
9	Fouli, Yvonne	Greenhouse Gt The School of f	10.11575/supp	https://www.pro	Despite numerous national and international agreements, climate change remains a global challenge. This abstract mentions farmers.		Yes, this abstract mentions farmers.	
10	Johannes Mey	Impact of Differ Horticulture	10.3390/hortcu	https://www.pro	Effective and efficient nutrient management is essential for sustainable agriculture. This abstract mentions farmers.		Yes, this abstract mentions farmers.	
11	Blanco Julien	A novel ecosys Sustainability	10.1007/s11621	https://www.pro	The ecosystem service framework has been widely used in the assessment of ecosystem services. This abstract mentions farmers.		Yes, this abstract mentions farmers.	
12	Tebbutt, Charlie	Participatory m People and Nat	10.1002/pan3.1	https://www.pro	Wildfires have increased in protected areas worldwide. This abstract mentions farmers.		Yes, this abstract mentions farmers.	
13	Alexandre Gori	Improving prod PLoS One	10.1371/journal	https://www.pro	About the Authors: Alexandre Gori Main Roles Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Supervision, Validation, Visualization, Writing—original draft, Writing—review & editing. This abstract mentions farmers.		Yes, this abstract mentions farmers.	
14	Bibi, Sofia	Trial Agring agri-en Environmental	10.1088/1748-6	https://www.pro	The global recognition of modern agricultural practices' impact on the environment has led to the development of agroecology. This abstract mentions farmers.		Yes, this abstract mentions farmers.	
15	Heywood, Sarah	Revealing stak Conservation S	10.1111/iscp2.31	https://www.pro	Biodiversity conservation in protected areas requires strict legal limitations to land use. In the Civilian Control Zone (CCZ) of the Republic of Korea (ROK), military control has been a major barrier to biodiversity conservation. This abstract mentions farmers.		Yes, this abstract mentions farmers.	
16	Gupta, M M	AI Exploring econ Symbiosis	10.1007/s13191	https://www.pro	The arbuscular mycorrhizal (AM) symbiosis has long been projected as one of the sustainable saviors for achieving food security for mankind. However, there exist conflicting views on the role of AM in sustainable agriculture. This abstract mentions farmers.		Yes, this abstract mentions farmers.	
17	Lutz C Pinheiro	Voinat National. Annals	10.3390/an1111	https://www.pro	Simple Summary To meet current global food supply challenges, researchers and technicians are searching for alternative livestock systems that promote highly sustainable and resilient production. This abstract mentions farmers.		Yes, this abstract mentions farmers.	
18	Muralikrishnan, Elucidating Trai Land		10.3390/land10	https://www.pro	Rice is grown under diverse agro-climatic conditions and crop management regimes across the globe. Emerging climatic vulnerabilities and the mismatched farm practices have led to the degradation of rice-based systems. This abstract mentions farmers.		Yes, this abstract mentions farmers.	
19	Srinellon, Elsal Do	Digital Cimi Agriculture	10.3390/agricul	https://www.pro	Digital climate services can support agricultural management decisions under uncertain climatological conditions and may contribute to achieving the ambitions of the fourth agricultural revolution. This abstract mentions farmers.		Yes, this abstract mentions farmers.	
20	Berenghini, An	Vineyard Yield Agronomy	10.3390/agronc	https://www.pro	Purpose—knowing in advance vineyard yield is a critical success factor for growers and winemakers can achieve the best balance between vegetative and reproductive growth. This abstract mentions farmers.		Yes, this abstract mentions farmers.	
21	Hanna Górka	Factors Limiting Agriculture	10.3390/agricul	https://www.pro	Our study aimed to explore the factors limiting the development of the organic food sector in Poland from the perspective of processing, distributing, and retailing companies. This abstract mentions farmers.		Yes, this abstract mentions farmers.	
22	Chirali, Eric	H Remote Sent Remote Sensin	10.3390/rs1311	https://www.pro	Monitoring biodiversity on a global scale is a major challenge for biodiversity conservation. Field assessments commonly used to assess patterns of biodiversity and habitats are labor-intensive and costly. This abstract mentions farmers.		Yes, this abstract mentions farmers.	
23		Sustainable Loi Sustainability	10.3390/su1301	https://www.pro	The Jamelista autochthonous bovine breed has a sustainable production and is part of the culture of the Portuguese territory, representing a touristic attraction and origins.		Yes, this abstract mentions farmers.	

FIGURE 4: Applying GPT for Google Sheets and Docs formula for remaining records.

R14 `=GPT("write yes if this abstract mentions farmer or land manager",J2)`

	J	K	L	M	N	O	P	Q	R
1	Abstract Note	farmers/ land managers	Carbon market	Agritourism	Ecosystem	Biodiversit	Renewable Ener	Climate change	Included?
2	Purpose—knowing in advance vineyard yield is a critical success factor for growers and winemakers can achieve the best balance between vegetative and reproductive growth.	Yes, this abstract mentions farmers and their practical applicability.	No	No	No	No	No	No	No
3	Our study aimed to explore the factors limiting the development of the organic food sector in Poland from the perspective of processing, distributing, and retailing companies.	No	No	No	No	No	No	No	No
4	The Jamelista autochthonous bovine breed has a sustainable production. Yes, this abstract mentions rural and livestock population evolution and	No	No	No	No	No	No	No	No
5	Purpose This paper investigates the incentives to coffee farmers to participate	No	No	No	No	No	No	No	No
6	Land related inequality is a central component of the wider inequality that I Yes, this abstract mentions farmers and indigenous people as being affe	No	No	No	No	No	No	No	No
7	SUMMARY The diversity of farms and farming systems in sub-Saharan Yes. The abstract mentions farmers and other stakeholders.	No	No	No	No	No	No	No	No
8	Harmonia axyridis is a global invasive alien species and its ecological effe Yes, this abstract mentions farmers and advisers as local stakeholders.	No	No	No	No	No	No	No	No
9	Purpose Polymakers and stakeholders lack empirical evidence on the eff Yes, this abstract mentions farmers.	No	No	No	No	No	No	No	No
10	Purpose This study forms part of a larger project funded by Cariplo Founde	No	No	No	No	No	No	No	No
11	Land area devoted to organic agriculture has increased steadily over the l Yes, this abstract mentions farmers and land managers.	No	No	No	No	No	No	No	No
12	Purpose The purpose of this paper is to determine the factors influencing a	No	No	No	No	No	No	No	No
13	The policy implications for success in the next stage Grain-for-Green Proje Yes, this abstract mentions farmers.	No	No	No	No	No	No	No	No
14	Purpose The purpose of this paper is to assess the sustainability Yes	No	No	No	No	No	No	No	No
15	(Table presented) Summary Arbuscular mycorrhizal fungi (AMF) are use Yes, this abstract mentions farmers.	No	No	No	No	No	No	No	No
16	The development of oil palm plantations has pushed for massive land-use Yes, this abstract mentions farmers and their livelihoods being impacted	No	No	No	No	No	No	No	No
17	Meeting the nutritional needs of a dynamically developing global society is No	No	No	No	No	No	No	No	No
18	This research aims to determine the motivation for the community-based i Yes, this abstract mentions farmers and agricultural product entrepreneurs.	No	Yes	Yes, this abt	No	No	No	No	Yes
19	Sustainable intensification (SI) of low input farming systems is promoted a Yes, this abstract mentions farmers.	No	No	Yes, this abt	Yes, this	No	Yes	Yes, this abstr	Yes
20	The food systems and territories of Indigenous Peoples sustain much of it Yes, this abstract mentions farmers or land managers indirectly through	No	No	Yes, this abt	Yes	No	Yes	Yes	Yes
21	Smallholder farming systems in southern Africa are characterized by low-l Yes, this abstract mentions farmers and land managers.	No	No	Yes, this abt	Yes, this	No	Yes, this abstr	Yes, this abstr	Yes
22	While Ethiopia and South Sudan are the native habitats for Coffea arabica Yes, this abstract mentions farmers.	No	No	No	Yes	No	No	Yes	Yes
23	Agroforestry has been practiced for decades and is undoubtedly an import Yes, this abstract mentions farmers.	Yes, the abstract m	No	Yes, this abt	No	Yes, the abstract	No	Yes	Yes
24	Conventional tillage (CT)-based agriculture is known to be ecologically ind Yes, this abstract m	No	Yes, this abt	No	No	No	Yes	Yes	Yes
25	Despite numerous national and international climate conferences, meeting Yes, this abstract mentions farmers and producers.	Yes, the abstract m	No	No	No	No	Yes, this abstr	Yes	Yes
26	Effective and efficient nutrient management is central to best-practice agri Yes, this abstract mentions farmers.	No	No	No	Yes, this	No	Yes	Yes	Yes
27	The ecosystem service framework has been instrumental in navigating loc Yes, this abstract mentions farmers.	No	No	Yes, this abt	Yes, this	No	No	Yes	Yes
28	Wildfires have increased in protected areas (PAs) of the Colombian Amazi Yes	No	No	Yes, this abt	Yes	No	Yes, this abstr	Yes	Yes
29	About the Authors: Alexandre Gori Main Roles Conceptualization, Data cu Yes, this abstract mentions family farmers and smallholder livestock and	No	No	Yes, this abt	Yes, this	No	Yes, this abstr	Yes	Yes
30	The global recognition of modern agricultural practices' impact on the env Yes, this abstract mentions farmers and land managers.	No	No	Yes, this abt	Yes, this	No	No	Yes	Yes
31	Biodiversity conservation in protected areas requires strict legal limitations Yes, the abstract mentions farmers.	No	No	No	Yes, this	No, this abstr	No	Yes	Yes
32	The arbuscular mycorrhizal (AM) symbiosis has long been projected as on Yes, this abstract mentions farmers.	No	No	Yes, this abt	Yes	No	Yes, this abstr	Yes	Yes

FIGURE 5: Assessing record relevance based on single scans.

Upon rechecking, we found that GPT for Google Sheets and Docs sometimes failed to deliver the requested outputs. For example, it incorrectly returned a “no” for an abstract containing the term “agritourism market.” However, GPT

for Google Sheets and Docs does possess some level of understanding of concepts, as demonstrated by its response to our request.

$$= \text{GPT} \left(\begin{array}{l} \text{"write yes if the abstract mentions any type of environmental markets",} \\ \text{[Abstract cell reference]} \end{array} \right)$$

Although GPT for Google Sheets and Docs returned relatively good results (e.g., identifying many types of environmental markets appearing in the abstracts), there were instances where it identified types of markets that were not environmental markets. For example, it classified “agri-environmental subsidies” as an environmental market, which is inaccurate.

The third caution is that if we apply the same GPT for Google Sheets and Docs formula to a large number of records at once, the likelihood of GPT for Google Sheets and Docs making mistakes increases. As such, we recom-

mend running it on a small number of records for better control.

The fourth caution is that after obtaining the screening outputs from your formulas, it is recommended to copy and paste them as values only. This will remove the GPT for Google Sheets and Docs functions (i.e., =gpt()). Failure to do so will result in ChatGPT running the formulas again every time columns are inserted or moved around.

Below is a summary outlining the advantages and disadvantages of using GPT for Google Sheets and Docs at this particular stage.

Strengths

- GPT for Sheets and Docs can accelerate the screening of keywords in research.
- The most significant benefit is that GPT for Sheets and Docs can quickly apply the screening results from the first instance to the remaining records.
- In some cases, GPT for Sheets and Docs shows an ability to comprehend research concepts.

Weaknesses

- GPT for Sheets and Docs is currently better than keyword counting, but it has not yet reached the level of sufficiently understanding a research concept. Therefore, researchers need to experiment and check the formulas to find the best ones for their purposes.
- GPT for Sheets and Docs can occasionally make mistakes, which requires researchers to cross-check the outputs.
- GPT for Sheets and Docs automatically reruns its formulas, which can result in the loss of obtained data or extra fees.

Box 2: Advantages and disadvantages of using GPT for Google Sheets and Docs in screening abstracts.



FIGURE 6: Using Casper AI to assess online full-text records.

3.2.2. Full-Text Screening. After the title and abstract screening, 63 studies were identified as relevant for full-text screening. There are several options for full-text screening, depending on how the full-text can be accessed. At the time of the experiment, the usefulness of AI-powered tools is somewhat limited due to their lack of a rich understanding of the research objectives, concepts, and context. However, it is still possible to take advantage of how the available tools can assist in the semantic search of full-text. Semantic search is a technique that seeks to understand the meaning and context behind search queries.

Instead of searching only for the keywords “carbon market” and “climate outcomes,” a semantic search can identify sections where the text discusses the relationship between the two concepts. Using semantic search powered by LLMs such as ChatGPT, researchers can more effectively identify relevant sections in the full-text documents and better understand the context in which the keywords and related

concepts are discussed. This, in turn, can lead to a more comprehensive and accurate assessment of the available evidence during the screening process.

The tool to use will depend somewhat on the availability of full-text forms. At the time of our experiment, Casper AI was only capable of scanning web pages, while ChatPDF could read and provided information based on PDFs.

3.2.3. Full-Text Available Online as Web Page Using Casper AI. Casper AI is a Google Chrome-based extension designed to help users summarize the content of web pages. It is particularly useful when documents are available as full-text web pages, which means Casper AI can access and process these texts directly.

We activated Casper AI and visited the website that contained the full-text of the record we wished to screen. Two useful ways to interact with Casper AI (see Figure 6) include the following:

TABLE 4: Using Casper AI in querying research articles.

Role	Text	Commentary
User	Does the article mention any type of environmental market?	Query to assess relevance of full-text
Casper	The article does not mention any specific type of environmental market, but it does emphasize the importance of environmental conservation as one of the dimensions of community-based agritourism. The study suggests that community-based agritourism can help to maintain local cultural and biological diversity through ecotourism management.	Casper AI stated that it did not identify any types of environmental market but highlighted an emphasis on related topics for our consideration.

TABLE 5: Using ChatPDF in querying research articles.

Role	Text	Commentary
User	Does the article mention any type of environmental market?	Query to assess relevance of full-text
ChatPDF	I'm sorry, but I could not find any mention of an environmental market in the pages of the PDF that you provided. The article mainly focuses on the development of a novel ecosystem service cascade model to navigate sustainability problems in a changing agricultural landscape in Brazil. It discusses the interactions between ecosystem services and ecosystem disservices and how they can be taken into account to support sustainability.	ChatPDF indicated that the article was not relevant. The second and third sentences provided extra information for our consideration.

- Using prebuilt functions such as “summarize this page”
- Asking specific questions to understand the content of the full-text.

Table 4 presents an example of an exchange with Casper AI in assessing the relevance of the record. This information aided us in assessing the article’s relevance. Although Casper AI did not identify any specific type of environmental market, it was helpful to cross-check the article. After a more detailed read, the research team decided that agritourism could be classified as a type of environmental market. This decision demonstrates that while AI-powered tools can serve as valuable assistants, the ultimate responsibility for making decisions regarding the inclusion and exclusion criteria rests with the researchers. Such decisions should reflect the research purpose, the theoretical framework, and existing knowledge. Consequently, we advanced this article to the next stage of data extraction.

3.2.4. *Full-Text Available as PDF Only.* If access to the full-text web page is not possible, an alternative is to obtain the full-text as a PDF and use one of two platforms to apply AI-powered tools to assess its relevance for a rapid review. ChatPDF allows the uploading of PDFs to its website, where you can then ask it questions about the content and conduct a semantic search. At the time of the experiment, Perplexity, SciSpace and Elicit are alternative AI tools that can search literature, summarize research articles, and allow users to deep dive into their PDFs through a chat-based interface. Within the scope of this article, we only discuss ChatPDF.

3.2.4.1. *Using ChatPDF.* Here, we present ChatPDF’s two responses to our question: “Does the article mention any type of environmental market?” In the first case (Table 5),

upon its scan, ChatPDF reported two types of information: First, it did not find any mention of the concept “environmental market.” Second, it also identified the main topic that the article focused on. This second piece of information helped us assess ChatPDF’s credibility in our further read. To verify its output, we carefully reviewed the article to ensure the accuracy of the AI-generated findings. It is crucial to cross-check the outcomes produced by AI-powered tools to maintain the quality of our research. Upon closer examination, we confirmed that the article focused on a “novel ecosystem service cascade model” and did not discuss the specific types of environmental markets. Consequently, we decided to exclude the article from our analysis. In the second case (Table 6), ChatPDF was able to identify that the article discussed carbon markets, which was relevant to our research topic. Again, we further checked ChatPDF’s outputs by requesting specific quotations from the article text (second query) so that we could read the most relevant sections of the article. ChatPDF’s capacity to extract exactly the quotations from the original text helped us build trust in this tool.

Below is a summary outlining the advantages and disadvantages of using Casper AI/ChatPDF at this particular stage.

4. Concluding Discussion

4.1. *Data Extraction and Quality Assessment (Step 6).* Although we have not yet used AI-powered tools to assist in data extraction and quality assessment, we believe that LLMs such as ChatGPT can help researchers extract relevant data from included studies more quickly and accurately, compared to manual extraction. Semantic search and identification of relevant information could save time and reduce the risk of human error during the data extraction process.

TABLE 6: Prompting with ChatPDF.

Role	Text	Commentary
User	Does the article mention any type of environmental market?	Query to assess relevance of full-text
ChatPDF	Yes, the article mentions carbon markets in relation to agricultural producers in Alberta. A study conducted by the University of Alberta in 2012 found that local markets and farm-specific factors affected participation in carbon markets more than global or provincial benefits (Page 9). However, the article does not provide an extensive discussion of environmental markets beyond this example.	ChatPDF indicated that the article was relevant with a detailed information.
User	So, please give some quotations on the impact of environmental market on GHG emissions	Query to request specific quotations
ChatPDF	Certainly! Here are some quotations from the article on the impact of environmental markets on GHG emissions - “Mechanisms such as setting a carbon price, cap-and-trade systems and tax-based policies contribute to decreased GHG emissions.” (Page 4) - “Agriculture has the potential to mitigate GHG emissions by applying BMPs that reduce emissions and increase carbon storage in soils.” (Page 4) - “Participation in environmental markets can provide financial incentives for land managers, farmers, and landholders to adopt agricultural land management practices that reduce GHG emissions and increase carbon storage in soils.” (Not a direct quotation from the article, but a summary of its content based on information from Pages 4 and 13)	ChatPDF was able to provide original quotations to back up its scanning result.

Strengths

- Casper AI/ChatPDF is helpful in summarizing the key findings of an article.
- Casper AI shows a better capacity to understand research concepts by identifying specific findings related to specific concepts or relationships between concepts.
- ChatPDF is more useful in identifying the pages that contain specific information.

Weaknesses

- Casper AI/ChatPDF can sometimes misidentify information from an article, so it is better to narrow down to specific sections of the article to enhance its scanning capacity.
- Like GPT for Sheets and Docs, both tools have not yet reached the level of understanding specific academic concepts. As a result, researchers need to spend time interacting with these tools through clear prompts to enhance their search capabilities.

Box 3: Advantages and disadvantages of using Casper AI and ChatPDF.

These tools could apply quality assessment checklists or guidelines, such as Cochrane’s risk of bias assessment guideline [32], to determine a score for each included article. This could help researchers more accurately assess the credibility of the evidence and make informed decisions about the inclusion or exclusion of studies.

However, the tendency of these tools to confabulate data or become confused about what information is in the document (accurate and relevant) and what information is in its training dataset (inaccurate and misleading) is a cause for concern.

4.2. *Knowledge Synthesis (Step 7).* AI-powered tools have shown their capacities for analyzing large sets of qualitative information and generating summaries, codes, and themes [6–8]. Applying ChatGPT in Template Analysis, Nguyen-Trung [33] found that under researchers’ thorough methodological guidance, supervision, and decision making, generative AI can play the role of a research assistant in data

analysis. As such, one potential area to apply LLMs is helping researchers identify patterns, trends, and relationships across the included articles [17]. While it is time-consuming to reanalyze text data to assess the effectiveness of an intervention on another new outcome, it is likely that AI-powered tools will make replication and extension of existing review methods easier. Effect size extraction for meta-analysis or other aggregation, such as vote counting, could also be assisted.

4.3. *Reporting and Disseminating Findings (Step 8).* Beyond clearly written summaries of findings for inclusion in reports, articles, briefings, and other written materials, LLMs could be applied to assist readers and stakeholders, who have a language barrier or no scientific background or tools in interpreting the findings of research. For example, creating a PDF of the evidence review and uploading it to ChatPDF can help readers explore the findings of the work and apply them to their own context in cases where the research team

does not have resources and time. ChatGPT can help researchers adapt their writings into more user-friendly products with fewer jargon, less formal language, and more local language phrases. This is also a highly scalable approach to research dissemination, because it allows research users to engage with the findings on their own terms, empowering them to ask questions, seek clarifications, and apply the findings to their context. By streamlining and enriching the reporting and dissemination processes, researchers can more effectively share their insights and contribute to the broader scientific community.

4.4. Implications. In this article, we show that the integration of AI-powered tools, such as ChatGPT, GPT for Google Sheets and Docs, and ChatPDF proved to be a valuable addition to the evidence review and synthesis process. Our article adds to the current exploration of utilizing machine learning and AI tools specifically in screening titles and abstracts and more broadly in evidence review [13, 34]. We argue that these tools offer various benefits for researchers, such as the capacity to generate Boolean queries and carry out titles/abstracts and full-text screening, which can potentially match human performance [18, 35]. These benefits include reducing the time, effort, and required number of researchers to conduct evidence reviews [15].

However, they each have weaknesses, such as inconsistent results and occasional errors, and the need for back-and-forth interaction and clear prompts to enhance search capabilities. Furthermore, although these AI-powered tools such as ChatGPT, GPT for Google Sheets, and Docs demonstrate some comprehension of research concepts, they may still misinterpret materials or generate misleading descriptions or summaries of those concepts. This could be because that the AI models behind those tools were not specifically trained on the dataset that helps them capable of “understanding” specialised knowledge (e.g., in our experiments, the knowledge domain is experimental markets and behavioral science). This aligns with the efforts of other researchers who have highlighted the problems of inaccuracies, confabulation, plagiarism, and confusion that could negatively affect research quality, transparency, and human autonomy [16, 36]. Hence, these tools should not be treated as autonomous researchers who can act and think for themselves. AI can handle labor-demanding tasks for human researchers, but decision-making and interpretation of research findings should never be outsourced to AI. We advise researchers to exercise caution when using AI-powered technologies and to design a thorough research strategy to ensure effective monitoring and quality control. To improve their search capabilities and outcomes, each research team should invest time in connecting with them, testing appropriate prompts, and routinely fact-checking. Cross-checking among human researchers is critical to preventing individual researchers from being “misled by false or biased information” produced by ChatGPT or AI-powered tools [36].

In their examination of computer technologies/intelligence over 20 years ago, Salomon et al. already developed the idea of “partnership with an intelligent tool”, pointing out that this partnership transformed our perception of

‘ability’ from just about the capacity of a person’s mind to the performance of a mind-machine collaboration [37]. Today, despite their limitations in grasping concept meaning and research tasks, ChatGPT and other AI-powered tools have a significant potential to transform the entire research process in general and evidence review and synthesis in particular. LLMs hold promise for enhancing data extraction, quality assessment, and knowledge synthesis in evidence reviews. They can also improve research communication and dissemination through quick summaries and interactive platforms, removing language barriers and empowering readers to engage with findings, as well as foster efficient knowledge sharing within the scientific and nonscientific communities. By embracing these technologies, researchers can enhance their productivity, contribute to more robust and comprehensive evidence synthesis, and ultimately drive positive social change and impact on the world’s most pressing problems at speed and scale.

In short, the COVID-19 pandemic has created an unprecedented demand for synthesised knowledge for policymaking [13, 14], the deployment of ChatGPT and other AI-powered tools could be a timely support to address these demands. As the field of AI continues to advance with the development of customised AI models tailored to specific research tasks, we anticipate that human-AI partnerships will become increasingly prevalent in research and academia. To keep up with this inconceivable pace, research institutions and academics around the world will be virtually compelled to incorporate AI-powered technologies into their own organisational research and processes. Despite the promise of generative AI, we must also consider and act to prevent the exacerbation of disparities between researchers in developed and developing countries [36].

Data Availability Statement

The data is confidential due to the terms of reference for our project.

Disclosure

Please note that the preprint of this article [38] has been available on the OSF Preprint server at <https://osf.io/preprints/osf/pcrqf>.

Conflicts of Interest

The authors declare no conflicts of interest.

Author Contributions

Author contributions are presented according to the International Council for Medical Journal Editors criteria for authorship. Contribution to conception or design of the research: K.N.T. and S.K. Contribution to the acquisition of the research: K.N.T. and S.K. Contributed to analysis or interpretation of data for the research: K.N.T. and A.K.S. Drafted the research or revisited the research critically for important intellectual content: K.N.T., A.K.S.,

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. (*Supporting Information*)

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